THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** hot to cold	Always.
2	**(c)** sinks	Heavier.
3	**(d)** solid	Solid.
4	**(c)** radiation	Radiation.
5	**(b)** in the day	From sea to land.
6	**(c)** fluid	Liquid or gas.
7	**(c)** conduction	Solid.
8	**(a)** at night	From land to sea.
9	**(c)** waves	Rays.
10	**(d)** convection	Fluid movement.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Conduction (1); Convection and radiation (1).	1+1
12	Iron (1); Copper (1).	1+1
13	Wind from sea (1); During the day (1).	1+1
14	Heat transfer in solids (1); Particle to particle (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Spoon in hot tea (1); Iron rod in fire (1); Hot pan handle (1).	1+1+1
16	Cotton shirt (1); Light colours (1); Loose clothes (1).	1+1+1
17	Double walls (1); Vacuum (1); Cork (1).	1+1+1
	OR: Boiling water (1); Sea breeze (1); Room heater (1).	1+1+1
18	Sweater (1); Woollen cap (1); Jacket (1).	1+1+1
	OR: Earth (1); Water (1); Air (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heat moves in a solid (1); Spoon in hot tea (1); Particle to particle (1); Metals conduct (1); Wood does not (1).	1+1+1+1+1
	OR: Hot tea (1); Cold drinks (1); Soup (1); Travel (1); Picnics (1).	1+1+1+1+1
20	Fluid moves (1); Hot water rises (1); Cool water sinks (1); Currents form (1); Boiling water (1).	1+1+1+1+1
	OR: Light colours in summer (1); Dark in winter (1); Cotton in summer (1); Wool in winter (1); Loose clothing (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
- Writing that dark colours are better in summer; **light** colours reflect more radiation.
- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** more heat	Absorb.
2	**(d)** radiation	Rays.
3	**(c)** iron	Metal.
4	**(b)** reflect heat	Reflect.
5	**(d)** top	Cold air sinks.
6	**(a)** wood	Poor conductor.
7	**(d)** good conductors	Conduct.
8	**(d)** near the floor	Hot air rises.
9	**(c)** winter	Warm.
10	**(b)** an insulator	Poor conductor.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wood (1); Plastic (1).	1+1
12	Wind from land (1); At night (1).	1+1
13	Heat transfer in fluids (1); By moving fluid (1).	1+1
14	Reflect heat (1); Keep cool (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sun (1); Fire (1); Bulb (1).	1+1+1
16	Metal conducts (1); Heat moves along (1); Wood does not (1).	1+1+1
17	Trap air (1); Keep warm (1); Insulation (1).	1+1+1
	OR: Iron (1); Copper (1); Aluminium (1).	1+1+1
18	Cold air sinks (1); Cools the fridge (1); Convection (1).	1+1+1
	OR: No medium (1); Travels through space (1); As waves (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	No medium (1); Sun to Earth (1); Waves (1); Fire warms us (1); Dark surfaces absorb (1).	1+1+1+1+1
	OR: Poor conductor (1); Prevents burns (1); Safe (1); Heat does not pass quickly (1); Use tongs too (1).	1+1+1+1+1
20	Iron (1); Copper (1); Wood (1); Plastic (1); Rubber (1).	1+1+1+1+1
	OR: Cooking (1); Heating (1); Drying (1); Ironing (1); Industry (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
- Writing that dark colours are better in summer; **light** colours reflect more radiation.
- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** thermos flask	To reflect heat.
2	**(a)** summer	Reflect heat.
3	**(b)** metal	Conduct.
4	**(b)** lighter	Less dense.
5	**(b)** hot or cold	Insulated.
6	**(b)** plastic or wood	Insulator.
7	**(a)** energy	Energy.
8	**(d)** rises	Less dense.
9	**(b)** air	Air is an insulator.
10	**(a)** poor conductor	Insulator.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Glass is a poor conductor (1); Keeps heat in (1).	1+1
12	Heat transfer by waves (1); No medium needed (1).	1+1
13	Traps air (1); Keeps warm (1).	1+1
14	From the Sun by radiation (1); Through space (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wood (1); Plastic (1); Air (1).	1+1+1
16	Do not conduct (1); Safe to hold (1); Do not burn (1).	1+1+1
17	Use a cloth (1); Use tongs (1); Do not touch (1).	1+1+1
	OR: Handles (1); Quilts (1); Flasks (1).	1+1+1
18	Hot air rises (1); Convection (1); Lighter than cold air (1).	1+1+1
	OR: Shade (1); Reflect heat (1); Keep cool (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day (1); Land warms faster (1); Warm air rises (1); Cool air from sea (1); Wind blows to land (1).	1+1+1+1+1
	OR: Conduct heat (1); Strong (1); Do not break (1); Last long (1); Easy to clean (1).	1+1+1+1+1
20	Night (1); Land cools faster (1); Sea warmer (1); Air over sea rises (1); Wind blows to sea (1).	1+1+1+1+1
	OR: Sea breeze (1); Land breeze (1); Boiling water (1); Rising smoke (1); Clouds (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
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- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
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- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

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